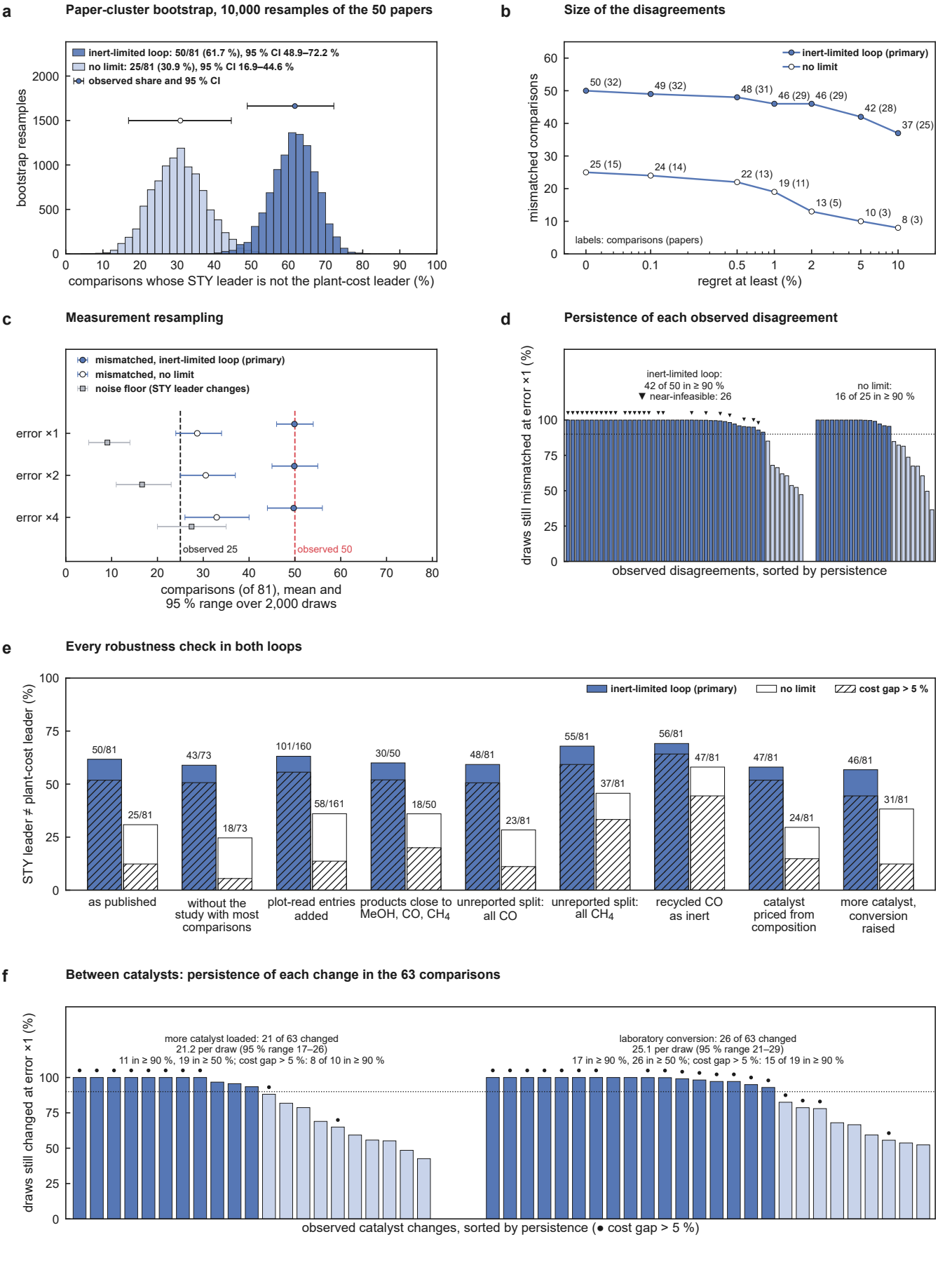

a
Paper-cluster bootstrap, 10,000 resamples of the 50 papers
inert-limited loop: 50/81 (61.7 %), 95 % CI 48.9–72.2 %
no limit: 25/81 (30.9 %), 95 % CI 16.9–44.6 %
observed share and 95 % CI
bootstrap resamples
comparisons whose STY leader is not the plant-cost leader (%)
b
Size of the disagreements
inert-limited loop (primary)
no limit
mismatched comparisons
regret at least (%)
labels: comparisons (papers)
c
Measurement resampling
mismatched, inert-limited loop (primary)
mismatched, no limit
noise floor (STY leader changes)
error ×1
error ×2
error ×4
observed 25
observed 50
comparisons (of 81), mean and 95 % range over 2,000 draws
d
Persistence of each observed disagreement
inert-limited loop:
42 of 50 in ≥ 90 %
▼ near-infeasible: 26
no limit:
16 of 25 in ≥ 90 %
draws still mismatched at error ×1 (%)
observed disagreements, sorted by persistence
e
Every robustness check in both loops
inert-limited loop (primary)
no limit
cost gap > 5 %
STY leader ≠ plant-cost leader (%)
50/81
25/81
43/73
18/73
101/160
58/161
30/50
18/50
48/81
23/81
55/81
37/81
56/81
47/81
47/81
24/81
46/81
31/81
as published
without the study with most comparisons
plot-read entries added
products close to MeOH, CO, CH4
unreported split: all CO
unreported split: all CH4
recycled CO as inert
catalyst priced from composition
more catalyst, conversion raised
f
Between catalysts: persistence of each change in the 63 comparisons
more catalyst loaded: 21 of 63 changed
21.2 per draw (95 % range 17–26)
11 in ≥ 90 %, 19 in ≥ 50 %; cost gap > 5 %: 8 of 10 in ≥ 90 %
laboratory conversion: 26 of 63 changed
25.1 per draw (95 % range 21–29)
17 in ≥ 90 %, 26 in ≥ 50 %; cost gap > 5 %: 15 of 19 in ≥ 90 %
draws still changed at error ×1 (%)
observed catalyst changes, sorted by persistence (● cost gap > 5 %)